

Exoplanet Atmosphere Assignment(Week 5):

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Q 1

(a)

Given:

$$L_* = 3.827 \times 10^{26} W$$

$$A = 0.3$$

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

For $T_{eq} = 273K$, we get:

$$a = \sqrt{\frac{(1-A)L_*}{16\pi\sigma T_{eq}^4}} = 0.86955069 \text{ A.U.}$$

For $T_{eq} = 373K$, we get:

$$a = \sqrt{\frac{(1-A)L_*}{16\pi\sigma T_{eq}^4}} = 0.46580327 \text{ A.U.}$$

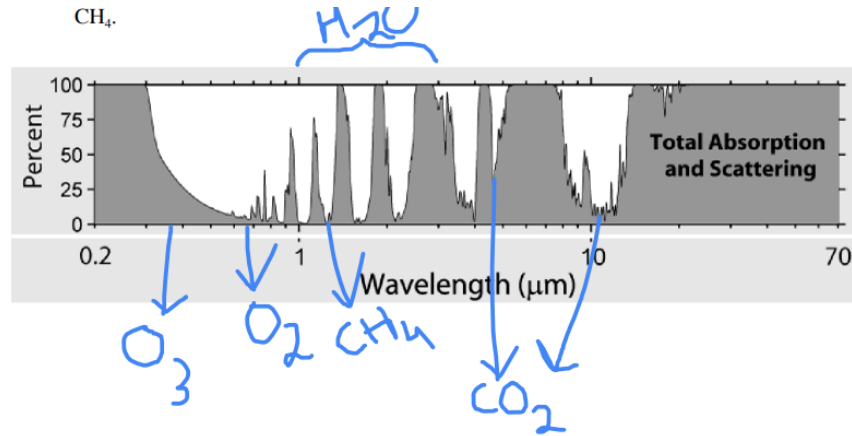
so the range of habitable zone is (0.46580327,0.86955069) A.U.

(b)

The toy model is an **underestimate** of the habitable zone.

One reason can be that we are not taking the **value of Albedo (A)** properly.

Q 2



Labelled absorption zones of H_2O , CO_2 , CH_4 , O_2 , O_3 .

Q 3

(a)

Given:

$$M = 4M_{\oplus}$$

$$r = 2r_{\oplus}$$

$$G = 6.673 \times 10^{-11} m^3 kg^{-1} s^{-2}$$

Calculating, we get,

$$v_e = \sqrt{\frac{2GM}{r}} = 15805.58 \text{ m/s} = 15.8 \text{ km/s}$$

(b)

Since the XUV influence is 10^2 times Earth and $V_e = 15.8$ km/s, we see that Terra II lies in a zone where it has lost its atmosphere.

